HARMONIZED CURRICULUM FOR MASTER OF SCIENCE (MSc) IN PHYSICS

CURRICULUM DESIGN TEAM:

Abebe Belay (PhD)

Adama Science and Technology University

Amanuel Fessahatsion (PhD)

Hawassa University

Chali Yadeta (MSc)

Jimma University

Girma Goro (PhD)

Haramaya University

Haileeyesus Workineh (MSc)

Bahir Dar University

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1. Introduction

Physics, as one of the fundamental sciences, is concerned with the observation, understanding and prediction of natural phenomena and the behavior of man-made systems. It deals with profound questions about the nature of the universe and with some of the most important practical, environmental and technological issues of our time. The scope of Physics is broad and encompasses mathematical and theoretical investigation, experimental observation, computing technique, technological application, material manipulation and information processing. Physics seeks simple explanations of physical phenomena based on universal principles stated in concise and powerful language of mathematics. The principles form a coherent unity, applicable to objects as diverse as DNA molecules, neutron stars, super-fluids, and liquid crystals. Findings in Physics have implications in all walks of life ranging from the way we perceive reality to gadgets of everyday use.

Ethiopian Universities have been training physicists and physics teachers for years. The programs have been revised and implemented at different times. Of these programs, the BEd programs implemented since the year 1996 were designed with emphasis on professional courses at the expense of subject area courses. As a result teachers in almost all preparatory schools of Ethiopia became deficient of subject matter area and challenged to properly deliver the course. The research conducted by the Ministry of Education has been identified the same and recommended the upgrading of these BEd teachers to MSc level with the modality of on job training through summer (Kiremt) program [MOE Document].

2. Justification

The launching of MSc program targeting the upgrading of teachers in Ethiopian preparatory schools will create opportunity to fill the deficiencies currently observed in their basic knowledge of the subject matter area.

According to the Ethiopian Education and Training Policy, preparatory schools shall be staffed with MSc holders [MOE 1994]. This program creates an opportunity to implement the policy.

The program will enhance the teaching skill of preparatory school teachers and also give them in-depth knowledge in subject matter area. As a result they will be able to teach in preparatory schools more effectively and efficiently.

3. The Master of Science (MSc) Program in Physics

Master of Science (MSc) in physics is a post graduate degree program designed so that the students admitted to the program, acquire an in- depth knowledge in diverse areas of physics at the most fundamental as well as at the advanced level and acquire advanced teaching methods of Physics. After the completion of the course, the students are expected to gain enough competence and proficiency in physics, which would enable them to pursue research in physics and teaching physics, to take up teaching jobs, and to get useful employment in any private or government institutions where knowledge of physics is an essential requirement.

3.1 Course Work

The program includes bridging courses to be completed before starting the MSc courses as a requirement. In the MSc program advanced physics courses and advanced teaching methodologies in physics are included. All courses are to be offered in a student centered manner so that the students get exposed to all the important aspects of physics, and acquire detailed knowledge of the front- line areas of physics. Seminar presentations, writing term papers, doing assignments, peer collaboration, developing teaching and research skills, will form an integral part of the course work which can be used as a means of continuous assessment.

3.2 Thesis Research /Graduate Project Work

One of the essential parts of the course is to train the students to develop an independent way of thinking about important physics- related problems, to analyze them and look for the best possible solutions. Towards that end, each student shall be engaged in thesis research in the areas related to teaching physics or pure physics. Students who prefer to specialize in a specific field of physics shall be given an option to do a project instead of thesis, accordingly the graduation requirements varies as indicated below for each area of specialization.

3.3 Objectives

The program has the following key objectives:

- To fill the gaps identified regarding knowledge and skill of teachers in the preparatory schools
- To upgrade the skills and theoretical background of teachers in the preparatory schools
- To expose teachers to new methods of teaching learning process in schools

- To impart knowledge in diverse fields of physics at the advanced level
- To produce qualified professionals in physics, capable of teaching at preparatory schools and tertiary level, pursuing research in front- line areas of physics.

3.4. Graduate Profile

The students, who successfully completed the MSc course in physics**,** are expected to have the following career prospects:

- ✓ They will be efficient and versatile in teaching physics at preparatory schools or tertiary level.
- ✓ They can pursue research in both theoretical and experimental physics in any national or international institutes, thereby contributing in the advancement of Physics.
- ✓ Having developed the skill and power of analytical and mathematical thinking, they would be assets to any government or private jobs which require an ability to track problems and find solutions

3.5 Admission Requirements

- Candidates should be a graduate with BEd/BSc degree in Physics
- Candidates must qualify in the written entrance examination administered by the department
- Candidates must meet the criteria set by the School of Graduate Studies (SGS) in addition to the written entrance examination administered by the department
- Candidates must take the bridging courses suggested in the curriculum. They must obtain a minimum CGPA of 2.00 with no "F" grade to start with MSc courses.

3.6 Duration of study

The modality of course delivery is face to face but in summer (Kiremt) and hence requires 4 consecutive summers for completion with possible extension by one summer with justifiable cases following the approval of graduate counsel at the department or college level.

3.7 Graduation Requirements

Minimum number of required course work to accomplish for the MSc degree:

3.7.1 With Thesis in Physics Education

- 24 Cr.Hrs of compulsory course work
- 3 Cr.Hrs elective course work
- 6 Cr.Hrs thesis in Physics Education
- Should score at least a "**satisfactory**" rating in the Thesis or as stipulated in the university senate legislation
- Course performance of CGPA greater than or equal to 3.00 with no more than one 'C' or as stipulated in the university senate legislation

3.7.2 With Thesis in Pure Physics (Science)

- 24 Cr.Hrs of compulsory course work
- 6 Cr.Hrs specialization courses in physics
- 6 Cr.Hrs Thesis in Pure Physics (Science)
- Should score at least a "satisfactory" in Thesis or as stipulated in the university senate legislation
- Course performance of CGPA greater than or equal to 3.00 with no more than one 'C' or as stipulated in the university senate legislation

3.7.3 With Graduate Project Pure Physics (Science)

- 24 Cr.Hrs of compulsory course work
- 3 Cr.Hrs elective course work
- 6 Cr.Hrs specialization courses in physics
- 3 Cr.Hrs graduate project work in Pure Physics (Science)
- Course performance of CGPA greater than or equal to 3.00 with no more than one 'C' or as stipulated in the university senate legislation
- A minimum of 'B' in graduate project work or as stipulated in the university senate legislation

3.8 Degree Nomenclature

Master of Science Degree in Physics

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4 Course Requirements

4.1 Course Coding

Subject matter courses in this program are commonly coded with a prefix ‘Phys’ followed by three digit numbers. The first digit of the three-digit numbers designates the class year (i.e., ‘5’ for year I and ‘6’ for year II). However, professional courses are coded as ‘PhEd’

The middle digits represent various branches of physics, such that

0 General Physics

1 Laboratory/Technical Courses

2 Statistical Physics

3 Classical Mechanics, Astronomy, Astro, Space, Plasma & Stellar Physics

4 Modern Physics, Quantum Mechanics

5 Solid State Physics, Electronics, Semiconductor Devices

6 Atmospheric, Environmental, Sustainable Source of Energy, Geophysics

7 Electrodynamics, Modern Optics, Laser Physics

8 Nuclear, Medical & Radiation Physics, Biophysics

9 Projects, Thesis

The last digits stand for semester in which the course is offered i.e.

“ODD” last digit courses are offered during the first semester

“EVEN” last digit courses are offered during the second semester

4.2 Compulsory Courses in Physics MSc Program

Course Code	Course Title	Credit hours
Phys 521	Statistical Mechanics	3
Phys 501	Mathematical Methods of Physics	3
Phys 531	Classical Mechanics	3
Phys 551	Classical Electrodynamics	3
Phys 542	Quantum Mechanics I	3
Phys 502 or Phys 512	Computational Physics or Advanced Physics Laboratory	3
Phys 691	Seminar in Physics	1
Phys 692	Research Methodology in physics	1
PhEd 603	Advanced Methods of Teaching Physics	2
PhEd 604	Secondary School Physics Curriculum and Research	2
Total compulsory credit hours		**24**

4.3 Elective courses in Physics MSc program

Course Code	Course Title	Credit hours
Phys 602	Computational physics II	3
Phys 611	Optics Laboratory	3
Phys 613	Science of Measurements	3
Phys 621	Statistical physics I	3
Phys 622	Statistical physics II	3
Phys 623	Advanced Statistical Physics	3
Phys 631	AstirophysicI	3
Phys632	AstrophysicsII	3
Phys 633	Relativistic Astrophysics	3
Phys 634	Stellar Equilibrium and Evolution	3
Phys 635	Space Physics I	3
Phys 636	Space Physics II	3
Phys 637	Compact Objects	3

Phys 638	Radar Systems I	3
Phys 639	Radar Systems II	3
Phys 641	Quantum Mechanics II	3
Phys 643	Quantum Field Theory	3
Phys 651	Solid state physics I	3
Phys 652	Solid state physics II	3
Phys 653	Physics of semiconductor Devices	3
Phys 654	Polymer Physics	3
Phys 655	Nanoscience & Technology	3
Phys 656	Principles of Condensed matter	3
Phys 657	Electronic Structure of Condensed matter	3
Phys 658	Advanced Condensed Matter Theory	3
Phys 659	Experimental Electronics	3
Phys 662	Generation of Alternative Sources of Energy	3
Phys.661	Environmental Physics	3
Phys 663	Atmospheric Physics	3
Phys 671	Quantum Optics I	3
Phys 672	Quantum Optics II	3
Phys 673	Optical Spectroscopic Techniques	3
Phys 674	Fiber optics and non-linear effects	3
Phys 675	Laser Physics I	3
Phys 676	Cavity Mode Atomic Dynamics	3
Phys 677	Resonance Florescence and Laser dynamics	3
Phys 678	Laser Physics II	3
Phys 679	Quantum States of Light	3
Phys 681	Nuclear Physics I	3
Phys 682	Nuclear Physics II	3
Phys 683	Atomic and molecular physics	3
Phys 684	Biological Physics/Biophysics	3
Phys 685	Radiological Physics I	3
Phys 686	Radiological Physics II	3
Phys 691	Seminar in Physics	1
Phys 695	Selected Topics	3
Phys 696	Graduate Project	3
Phys 751	Electronics I	3
Phys 752	Electronics II	3
Phys 753	Physical electronics I	3
Phys 754	Physical electronics II	3
Phys 751	Electronics I	3
Phys 752	Electronics II	3
PhEd 603	Advanced method of teaching Physics	2
PhEd 604	Secondary school physics curriculum & Research	2

In addition to the **Compulsory courses**

a) Students for MSc with thesis in **Physics Education** should take at least one more course (3 credit hours) from elective courses and Phys 699 (Thesis with 6 credit hours)

b) Students for MSc with thesis in **Pure physics (science)** should take at least two more courses of 3 credit hours each from elective courses in the respective field of specialization and Phys 699 (Thesis with 6 credit hours)

c) Students for MSc with non-thesis should take at least three elective courses and Phys 696 (Graduate Project) with 3 credit hours each.

4.4 Course schedule

Summer I (Pre-MSc Program)

Course Code	Course Title	Credit Hours
Phys 212	Experimental Physics	2
Phys 371	Modern Optics	3
Phys 382	Nuclear Physics	3
Phys 451	Solid State Physics	3
Total		11

Summer II – MSc Program

Course Code	Course Title	Credit Hours
Phys 521	Statistical Mechanics	3
Phys 531	Classical Mechanics	3
Phys 501	Mathematical Methods of Physics	3
PhEd 603	Advanced Teaching in subject Matter	2
Total		11

Summer III – MSc Program

Course Code	Course Title	Credit Hours
Phys 551	Classical Electrodynamics	3
Phys 502 or Phys 512	Computational Physics or Advanced Physics Laboratory	3
Phys 542	Quantum mechanics	3
PhEd 604	Secondary School Physics Curriculum and Research	2
Phys 692	Research Methodology	1
Total		**12**

Summer IV – MSc Program (Thesis on Physics education)

Course Code	Course Title	Credit Hours
Phys **xxx**	Elective course	3
Phys 691	Seminar in Physics	1
Phys 699	Thesis	6
Total		**10**

Summer IV – MSc Program (Thesis on Pure Physics (Science))

Course Code	Course Title	Credit Hours
Phys 691	Seminar in Physics	1
Phys XXX	Specialization course I	3
Phys XXX	Specialization course II	3
Phys 699	Thesis	6
Total		**13**

Summer IV – MSc Program (With Project Work)

Course Code	Course Title	Credit Hours
Phys XXX	Specialization course I	3
Phys XXX	Specialization course II	3
Phys XXX	Elective course I	3
Phys 691	Seminar in Physics	1
Phys 696	Graduate Project	3
Total		**13**

4.5 Course Descriptions

Phys 501 Mathematical Methods of Physics (3)

Vector calculus, Eigen value problems and orthonormal functions, Matrix theory and tensor analysis, Laplace and Fourier transforms, Partial differential and integral equations, Special functions, Complex variables.

References

1. Mathematical Methods for Physicists: George Arkfen (Academic Press).
2. Applied Mathematics for Engineers and Physicists: L. A. Pipe (McGraw Hill)
3. Mathematical Methods-Potter and Goldberg (Prentice Hall of fudia).
4. Elements of Group Theory for Physicists: A. W. Joshi (Wiley Eastern Ltd.)
5. Mathematics for Physicists Dennery & Kryzywicki
6. Complex Variables & Applications R. V. Churchill
7. Linear Vector Spaces R. R. Halmos
8. Theory of Finite Groups L. Jansen and M. Boon

Phys 502 Computational Physics (3)

Fundamental methods of computational physics and applications, Numerical algorithms, System of linear equations, differential equations, computer simulations with selected methods, Examples and projects on scientific applications.

References

1. Introduction to Numerical Methods T. R. McCalla

2. Numerical Methods that work F. S. Acton

3. An Introduction to Numerical Analysis K. E. Atkinson

4. Numerical Recipes W. H. Press et.al

Phys 512 Advanced Physics Laboratory (3)

Minimum of 8 setups on advanced topics in optics, radiation Physics, electric and magnetic phenomena.

Phys 521 Statistical Mechanics (3)

Statistical description of systems, The statistical matrix, Thermodynamic potentials, Canonical and grand canonical ensembles, Equipartition, Maxwell-Boltzmann, Bose-Einstein and Fermi-Dirac Statistics, The electron gas, Black-body radiation, Phase transitions.

References

1. Huang Statistical Mechanics

2. Reif : Fundamentals of Statistical and Thermodynamics Physics

3. Rice : Statistical mechanics and Thermal Physics

4. Kittle: Elementary statistical Mechanics

5. P. T. Landsberg (Ed.)Problems in Thermodynamics and Statistical Physics

Phys 531 Classical Mechanics (3)

Variational principles, Lagrangian equations, Conservation laws, Symmetry and invariance properties, Scattering, Rigid body motion, Hamilton's equations, Canonical transformations, Hamilton-Jacobi equations, Small Oscillations.

References

1. Goldstein- Classical Mechanics.
2. Landau.and Lifshitz - Classical Mechanics.
3. A.Raychoudhary - Classical Mechanics

Phys 542 Quantum Mechanics I (3)

Postulates of quantum mechanics, Expectation values, The Schrodinger and Heisenberg equations, The state function propagator, Angular momentum Eigen values and Eigen states, Addition of angular momentum, Orbital angular momentum Eigen functions, The hydrogen energy Eigen values and Eigen states, Time-independent perturbation theory, Spin-orbit coupling, The Zeeman effect, The WKB approximation, Time-dependent perturbation, The classical radiation field, The quantum radiation field, transition rates for stimulated and spontaneous emission, The lifetime of an excited atomic state.

References

1. Quantum Mechanics L. Schiff

2. Quantum Mechanics E. Merzbacher

3. Practical Quantum Mechanics S. Flugge

4. Quantum Mechanics Mathews and Venkatesan

5. Quantum Mechanics M. P. Khanna

6. Principles of Quantum Mechanics P. A. M. Dirac

Phys 551 Classical Electrodynamics (3)

Solution of Poisson's equation, Solution of a boundary value problem with Green's function, Vector potential for a circular current loop, Maxwell's equations, Coulomb and Lorentz gauges, Reflection and refraction of plane electromagnetic waves, Dispersion relations, Motion of a charged particle in electric and magnetic fields, Retarded potentials, Electric dipole radiation, Lienard-Wiechert potentials, Radiation by an accelerated charge, Bremsstrahlung, Covariant formulation of electrodynamics.

References

1. J.D. Jackson-.-Classical Electrodynamics
2. Panofsky and Philips Classical Electricity and Magnetism
3. Introduction to Electrodynamics-Gritfiths
4. Landau and Lifshitz--Classical Theory of Field
5. Landau and Lifshitz. .Electrodynamics of Continuous Media

Phys 602 Computational Physics II (3)

Numerical solutions of partial differential equation I, Partial Differential equation II, Partial Differential Equation III, Stochastic Methods, Special Functions and quadrature.

References

1. Tao Pang, An Introduction to Computational Physics,Cambridge University Press, (1997)
2. R. Fitzpatrick, Computational Physics: Computer based learning unit, University of Leads, (1996).
3. H Gould, et al, An Introduction to computer simulation methods: Application to Physical System, 2nd ed., (1995).
4. Fitzpatrick, Introduction to Computational Physics, University of Texas.

Phys 622 Statistical Physics II (3)

Master equation and Jump processes, Spatially distributed systems, Bistability and escape problems, Quantum Markov processes.

References

1. Huag : Statistical Mechanics

2. Reif : Fundamentals of Statistical and Thermodynamical Physics

3. M. Plischke and B. Bergesen Equilibrium statistical physics

4. S. K. Ma Modern theory of critical phenomena

5. L. E. Reichl A modern course in statistical physics

6. J. K. Bhattacharya Statistical Mechanics

Phys 631 Astrophysics I (3)

Historical introduction Special relativity, Currents and densities, The energy-momentum tensor, Relativistic hydrodynamics, The principle of equivalence, Gravitational forces, Relations between $g^{\mu\nu}$ *and* $\Gamma^{\lambda}_{\mu\nu}$,The Newtonian limit, Tensor analysis, The principle of general covariance, Covariant/contravariant vectors and tensors, Covariant differentiation and differentiation along a curve, p-forms and exterior derivatives, Effects of gravitation, Curvature, Uniqueness of the curvature tensor, Parallel transport, Properties of $R_{\lambda\mu\nu\kappa}$ The Bianchi identities, Geodesic derivation, Einstein's field equations, Coordinate conditions, Energy-momentum and angular momentum of gravitation.

Phys 632 Astrophysics II (3)

Test of Einstein’s theory ,unbound and and bound orbits ,the general isotropic metric ,The Schwarzschild metric, Other metrics, General equations of motion, Post-Newtonian mechanics, Post-Newtonian approximation, Particle and photon dynamics, multipole fields, Post-Newtonian hydrodynamics, Gravitational radiation, The weak-field approximation, Plane waves, Quadrupole radiation, Detection of gravitational radiation, Stellar equilibrium and collapse, Differential equations of stellar structure, Stability, Polytropes ,White dwarfs and neutron stars, Time-dependent spherically symmetric fields, Comoving coordinates, Gravitational collapse and black holes.

Phys 633   Relativistic Astrophysics (3)

Covariant physics, The principle of equivalence, Tensor analysis, Effects of gravitation, Einstein's field equations, Classical tests of Einstein's theory, Post-Newtonian celestial mechanics, Gravitational radiation.

References

1. Gravitation and Cosmology, John Wiley and Sons, by Steven Weinberg.
2. Gravitation, Freeman and Company, by C. W. Meisner, K. S. Thorne and J. A. Wheeler.

Phys 634   Stellar Equilibrium and Evolution (3)

Stellar equilibrium, The stability of stars, Relativistic equations of stellar equilibrium, Relativistic equations for rotating stars, the theory of cold white dwarfs, neutron stars, the mass defect, stability of neutron stars, Stellar evolution, Evolution of a star up to the loss of stability, Instability of massive stars with nuclear sources of energy, stability of stellar evolution, Supernova, Evolution of a star with mass greater than the Oppenheimer-Volkoff limit, Relativistic collapse.

References

- Gravitation and Cosmology, John Wiley and Sons, by Steven Weinberg.
- Gravitation, Freeman and Company, by C. W. Meisner, K. S. Thorne and J. A. Wheeler.

Phys 635   Space Physics I (3)

The Sun, The Solar wind and the Interplanetary magnetic field, The earth's magnetic field, The ionosphere, Currents in the ionosphere, The magnetosphere, The aurora, Equatorial ionosphere and electrojet.

References

1. Michael C. Kelley, The Earth's Ionosphere, Plasma Physics and Elctrodynamics, Elsevier Inc,
2. USA (2009)
3. Robert Shunk and Andrew Nagy, Ionospheres Physics, Plasma Physics and Chemistry, Cambridge University press, USA (2009)
4. Hans Volland, Atmospheric Electrodynamics, Springer, Germany (1984)
5. Berreke, Physics of the Upper Polar Atmosphere, John Wiley and Sons Ltd, England (1997)
6. Anita Akio and Tuomo Nygren, Ionospheric Physics module (761658S), University of Oulu, Finland (2008)

Phys 636 Space Physics II (3)

Instrumentation (magnetometers, ionosonde, radar, GPS) and data analysis methods in space physics (time series analysis techniques, including filtering, Fourier series and power spectra), numerical simulations of space plasma, Recent advances in space physics research and instrumentation.

References

1. Michael C. Kelley, The Earth's Ionosphere, Plasma Physics and Elctrodynamics, Elsevier Inc,
2. USA (2009)
3. Robert Shunk and Andrew Nagy, Ionospheres Physics, Plasma Physics and Chemistry, Cambridge University press, USA (2009)
4. Hans Volland, Atmospheric Electrodynamics, Springer, Germany (1984)
5. Berreke, Physics of the Upper Polar Atmosphere, John Wiley and Sons Ltd, England (1997)
6. Anita Akio and Tuomo Nygren, Ionospheric Physics module (761658S), University of Oulu, Finland (2008)

Phys 637 Compact Objects (3)

Cold equation of state below neutron drip, White dwarfs, The equilibrium and stability of fluid configurations, Cold equation of state above neutron drip, Neutron star models, Pulsars, Black holes.

References

- Black Holes, Neutron Stars and White Dwarfs, Wiley inter-science, by S. L. Shapiro, S. A. Teukolsky.
- Gravitation and Cosmology, John Wiley and Sons, by Steven Weinberg.

Phys 638 Radar Systems I (3)

Radar fundamentals, radar cross section, radar detection, radar waveform analysis, pulse compression, radar wave propagation, clutter, radar antennas, target tracking.

References

1. Bassem R. Mahafza, Radar Systems Analysis and Design Using MATLAB, Chapman and Hall/CRC, USA (2000)
2. Merrill I. S., Introduction to Radar Systems, McGraw-Hill Campanies, Inc, USA, (2002)

Phys 639 Radar Systems II (3)

Review of probability theory, filtering, Bayesian estimation techniques, the Kalman filter, non-linear filters, modified gain techniques, Wiener filters, adaptive methods.

References

1. Bassem R. Mahafza, Radar Systems Analysis and Design Using MATLAB, Chapman and Hall/CRC, USA (2000)
2. Merrill I. S., Introduction to Radar Systems, McGraw-Hill Campanies, Inc, USA, (2002)

Physics 641 Quantum Mechanics II (3)

The Klein-Gordon Equation, Lorentz invariance, Free particle solutions, Interaction of a Kelin-Gordon particle with an electromagnetic field, The solution of the Kelin-Gordon equation for a square well potential, The Dirac Equation, Properties of Dirac matrices, Invariance of the Dirac Equation under Lorentz transformation, Parity transformation and Charge conjugation, Free particle solutions, Energy spectrum of a Dirac particle in a one dimensional square well potential, Interaction of a Dirac particle with an electromagnetic field, Spin and magnetic moment of the electron, Spin-orbit coupling, The hydrogen atom spectrum, Nonrelativistic limit of the Dirac Equation

References

1. Ashok Das and A.C. Millissiones : Quantum Mechanics,A Modern approach.(Garden and Breach Science Publishers)
2. E. Merzbaker : Quantum Mechanics, Second Edition (John Wiley and sons)
3. Bjorken and Drell : Relativistic Quantum Mechanics (MGraw Hill)
4. J.J. Sakuri : Advanced Quantum Mechanics (John Wiley)
5. F. Mandal & G. Shaw, Quantum Field Theory (John Wiley)
6. J,M. Ziman, Elements of Advance Quantum Theory, (Cambridge University ress).

Phys 643 Quantum Field Theory I (3)

Lagrangian formulation of classical fields, Noether's theorem, The Klein-Gordon field, The Dirac field, The Maxwell field, Parity, time reversal, and charge conjugation, The S matrix expansion, Wick's theorem, Feynman rules, Second order QED processes, Radiative corrections.

References

1. F. Mandal & G. Shaw, Quantum Field Theory (John Wiley)
2. J,M. Ziman, Elements of Advance Quantum Theory, (Cambridge University Press).
3. J. Sakurai Advance Quantum Mechanics
4. Bjorken and Drell Relativistic Quantum Fields. Vols. I & II
5. Mandl Quantum Field Theory

6. Weinberg Quantum Theory of Fields. Vols. I & II

Phys 651 Solid State Physics I (3)

Bloch theorem, Kronig-Penney dispersion relation, Nearly free electron approximation, Tightly bound electron approximation and its applications, **k.p** method for obtaining **E-k** relation and its application to wide gap and narrow gap semiconductors, Construction of Brillouin Zone, de Haas-van Alphen effect and Fermi surface determination, Boltzmann equation and its solution in crossed E and B fields, Galvano-magnetic transport coefficients in degenerate semiconductors, Donor and acceptor states, Distribution function of electrons in donor and acceptor states, Evaluation of Fermi level and carrier concentration in n-type and p-type semiconductors.

References

1. Kittle : Introduction to Solid State Physics
2. Patterson: Solid State Physics
3. R. A.Levy : Principles of Solid State Physics
4. Mckelvy: Solid State and Semi-conductor Physics.
5. J. S. Blakemore Solid State Physics
6. J. Ziman Principles of the Theory of Solids

Phys 652 Solid State Physics II (3)

Surface effects, Thermionic emission, Schottky emission, Field emission, Classical and quantum theory of harmonic crystals, Anharmonic effects in crystals, Thermal expansion and Gruneisen parameter, Phonon dispersion relation in metals, Quantum well structures, Semiconductor lasers, Diamagnetism, Paramagnetism, Magnetic ordering, Magnetic structures, Ferro-magnetism, Ferri-magnetism, Magnons, Spin waves, Superconductivity, Meissner effect, Energy gap, London equation, BCS theory of superconductivity, Flux quantization in a superconducting ring, Josephson effect, Point defects and dislocations.

References

1. Introduction to Solid State Physics by Kittel.
2. Patterson: Solid State Physics
3. R. A.Levy : Principles of Solid State Physics

4. Mckelvy: Solid State and Semi-conductor Physics.
5. J. S. Blakemore Solid State Physics
6. J. Ziman Principles of the Theory of Solids

Phys 653 Physics of semiconductor Devices (3)

Crystal structure, Energy band, Impurities, Carrier statistics, Optical property,

Non-equilibrium phenomena, Basic equations for semiconductor, p-n junction, Bipolar transistor, Metal-Semiconductor contacts, Field effect transistor, Photonic Devices: LED and Semiconductor laser, Photo detectors, Solar cells

Phys 654 Polymer Physics (3)

Molecular weight distribution, Polymer transitions (T_g), Polymer synthesis, Chain structure and configuration, Stereochemistry and isomerism of repeating units, Copolymers, Determination of molecular weight and radii of Gyration mark-houwink relation, Solubility parameters and thermodynamics of mixing, Molecular motion and de Gennes reptation, Rubber elasticity and statistical thermodynamics, Introduction to semiconducting conjugated polymers.

Phys 655 Nanoscience and Technology (3)

Definition of nano, nanoscience and technology; nanomaterials; nanoparticles; nanopowders; properties of nanomaterials; synthesis techniques including pyrolytic synthesis, sol-gel, hydrothermal processing, surfactant and sacrificial templating; Review underlying physical concepts; model systems for the study of physics on the nano-scale including nano-scale confinement effects and Knudsen transport effects. Concepts of aggregates; concepts of fractals; scaling laws; characterization techniques of nanomaterials; applications of nano-structured inorganic, organic, ceramics and hybrid powders; applications including use as catalysis, chemical sensors, coatings, reinforcing fillers.

Referernces

A. Frank J. Owens, Charles P. Poole Jr, Introduction to Nanotechnology

B. C. Binns, Introduction to Nanoscience and Nanotechnology.

C. Any books on Nano science and nano Technology.

Phys 656 Principles of Condensed Matter (3)

Energy levels in periodic potential and Bloch theorem, Electrons in a weak periodic potential, Tight binding model, Pseudopotential method, and other methods of calculating band structure, Density functional theory, Hartree Fock theory, Transport theory, Defects in condensed media, Quantum many body theory, Interacting subsystems, Homogenous and inhomogeneous semiconductors.

References

- Principles of Condensed Matter Physics, P.M. Chaikin and T.C. Lubenskii, Cambridge University Press.
- Condensed Matter Physics, Michael P. Marder, John Willey and Sons.
- Solid State Physics, Neil W. Ashcroft, N. David Mermin, Thomson Learning.
- Ch. Kittel, Quantum Theory of Solids, 2nd revised Edition, N.Y., John Willey and Sons, 1987.
- Ch. Kittel, Introduction to Solid State Physics, 8th Edition, John Willey and Sons, 2005.

Phys 657 Electronic Structure of Condensed Matter (3)

Plasmons, Polaritons, Polarons and other elementary excitations, Optical processes and excitons, Dielectric properties of insulators, Diamagnetism, Paramagnetism, Ferromagnetism, Antiferromagnetism, Spin glasses, Spinwaves, Giant magnetoresistance, Diluted magnetic semconductors, Magnetic ordering, Theory of superconductivity and superfluidity, Introduction to Polymer Physics.

References

- Principles of Condensed Matter Physics, P.M. Chaikin and T.C. Lubenskii, Cambridge University Press.
- Condensed Matter Physics, Michael P. Marder, John Willey and Sons.
- Solid State Physics, Neil W. Ashcroft, N. David Mermin, Thomson Learning.
- Ch. Kittel, Quantum Theory of Solids, 2nd revised Edition, N.Y., John Willey and Sons, 1987.
- Ch. Kittel, Introduction to Solid State Physics, 8th Edition, John Willey and Sons, 2005.

Phys 658 Advanced Condensed Matter Theory (3)

Many-body theory for low dimensional systems, Nanosystems, Density of states of nanostructures, Single electron systems, Photonic crystals, Corrals, Quantum Hall effect, Nanophotonics, Heterostructures, Plasmonics, NEMS, Imaging techniques using STM and AFM, Nanodevices, Nanomachines and Mesoscopic systems, Nano polymer physics, Strongly correlated systems, Fermi and Bose Systems at ultra low temperatures.

References

1. The Nanomaterials by C.N.R. Rao, A. Muller, and A.K. Cheetham (Volume 1 and 2), Wiley – VCH (2004).
2. Mesoscopic Electronics in Solid State Nanostructures by Thomas Heinzel, Wiely – VCH (2007).
3. Photonic Crystals: Molding the Flow of light by J.D. Joannopoulos, S.G. Johnson, J.N. Winn and R.D. Meade, Princeton University Press (2008).
4. Solid State Physics, Neil W. Ashcroft and N. David Mermin, Thomson Learning.
5. C. Kittel, Introduction to Solid State Physics, 8th Edition, John Willey and Sons, 2005.
6. Nanotechnology Demystified by L. Williams and D.W. Adams, McGraw Hill (2007).

7. Metamaterials and Plasmonics: Fundamentals, Modeling and Applications by S. Zouhdi, Sihvola, A.P. Vinogradov, Springer (2008).

Phys 659 Experimental Electronics (3)

Mathematical operations, Inverting and non-inverting amplifiers, Converters, Comparators, Relaxation oscillator, RC-Oscillators, Active filters, Operational Amplifiers, Characteristics, Basic logic gates, Arithmetic circuits, Multiplexers and de-multiplexers, Counters and registers, Ring and Johnson counters, Phase-lock loop (PLL) characteristics, Log and Antilog amplifiers, Instrumentation amplifiers, Signal conditioning using sensors.

Phys 661 Environmental Physics

Identification of our current energy resources: Fossil fuels, methane hydrates, hydroelectric, wind, geothermal; Nuclear, solar and tidal energies; Energy Conversion Processes: The first and second laws of thermodynamics, The generation of heat from fuels, geothermal, nuclear fission, nuclear breeders, nuclear fusion, solar thermal processes; Heat pumps, refrigerators, internal combustion engines, turbine engines. Photovoltaics, magneto-hydrodynamics, battery development. Heat management with cogeneration, and waste heat disposal, Energy Utilization: Energy transmission, superconductivity, Efficient use of energy in industry, transport and heating, Environmental consequences of energy use: Global climate change, Tropospheric and stratospheric ozone, the physics of El Nino, Air and water pollution, Nuclear radiation, Heat and micro climate.

References

1. Environmental Physics- Egbert Boeker, Rienk Van Grondelle (2011)

2. Principles of Environmental Physics, John Lennox Mopnteita, M. H. Unsworth (2008)

3. An Introduction to Environmental Physics of Soil, Water and Watershed< C. W. Rose

Phys 662 Generation of Alternative Energy Sources (3)

Fossil Fuels (Petroleum, natural Gas, Coal, Propane, Butane, Methanol etc..), **Bioenergy** (solid Biomass, Biodiesel, Biogas, Ethanol etc.), **Geothermal Energy** (Definition, resource exploration, potential, applications, Power plants), **Hydrogen** (Producing hydrogen, transporting, storing, and future technology), **Nuclear Energy** (Overview, current future technology, benefits and drawbacks, environmental impact, societal impact), **Solar Energy** (Basics, Passive solar design, Photovoltaic cells, solar ponds and solar towers, solar energy potential), **Wind Energy** (How wind energy works, current future technology, benefits and draw backs, wind turbines, Potentials, challenges and obstacles), **Water Energy** (Hydropower, hydroelectricity), **Energy conservation and Efficiency, Possible Future Energy sources**

Phys 663 Atmospheric Physics

Atmosphere of other planets and their Equilibrium temperatures; Hydrostatic Equation; Atmospheric thermodynamics: Atmospheric dynamics: Numerical Modeling of Atmospheric State; and Introduction to remote Sounding of Atmosphere.

References

1. An Introduction to atmospheric physics-DG. Andrews
2. An Introduction to Dynamic meteorology – J. Holton
3. Meteorology today – Ahrens, C. Donald

Phys 671 Quantum Optics I (3)

The Schrodinger Equation and Stationary states, Spherically Symmetric potentials in 3 Dimensions, Rigid rotator, The Hydrogen atom, Time Independent problems, Non degenerate case, Degenerate case, Zeeman effect, Stark effect, Time dependent perturbation theory, Fermi’s Golden Rule, Adiabatic approximation, Sudden approximation, Angular momentum and parity, Plank’s radiation law and the Einstein coefficients, Theory of simple optical processes, Theory of chaotic light and coherence, Quantization of the radiation field, states of quantized radiation field, Photons and the electromagnetic field, photon optics. The quantum radiation field, The photon number state, Coherent and chaotic states, The P and the Q functions, The photon number and the photon count distributions, Quadrature variance and photon statistics of single mode and two-mode squeezed states, The master equation, The Fokker-Planck equation, and Quantum Langevin equations for a cavity mode coupled to a squeezed vacuum reservoir, The quantum regression theorem and input-output relation, Collapse and revival, The master equation for a two-level atom coupled to a squeezed vacuum.

References

1. F.Kassahun , Fundamentals of quantum optics,(Lulu 2008)
2. F. Mandal & G. Shaw, Quantum Field Theory (John Wiley)
3. J.M. Ziman, Elements of Advance Quantum Theory, (Cambridge University Press).
4. Baldwin, ntroduction to Quantum Optics
5. Louisell, Statistical Theory of Radiation
6. Mandel & Wolf, Coherence & Quantum Optics

Phys 672 Quantum Optics II (3)

Matrix representation, Symmetry, Schrodinger Heisenberg interaction picture, Dirac's bra and ket notations, Collisions of identical particles, The scattering of light by atoms, Theory of the Laser, Interaction of radiation with atoms, Collective atomic interactions, The linear light amplifier, The sqeezed state of light, some quantum effects in non-linear optics, theory, Born's approximation, partial wave analysis, separation of the equation, Energy levels in a Coulomb field, Dirac's relativistic equation, Dirac's equation for a central field, Spin angular momentum, Spin-orbit coupling, Negative energy states. Power spectrum, photon antibunching, and atomic dynamics in resonance fluorescence, Photon statistics, power spectrum, and spectrum of intensity fluctuations in a two-level laser, Photon statistics, squeezing spectrum, and spectrum of intensity-difference fluctuations in parametric oscillation, Quadrature variance, mean photon number, and squeezing spectrum in second harmonic generation, Power spectrum, quadrature variance, squeezing spectrum, and photon statistics in a three-level laser.

References

1. F.Kassahun , Fundamentals of quantum optics,(Lulu 2008)
2. F. Mandal & G. Shaw, Quantum Field Theory (John Wiley)
3. J,M. Ziman, Elements of Advance Quantum Theory, (Cambridge University Press).
4. Baldwin Introduction to Quantum Optics
5. Louisell Statistical Theory of Radiation
6. Mandel & Wolf Coherence & Quantum Optics

Phys 673 Optical Spectroscopic Techniques (3)

UV-VIS and fluorescence spectrophotometer, IR and Raman Spectroscopy, EPR and NMR spectroscopy, laser induced fluorescence spectroscopy, laser induced break down spectroscopy, and other spectroscopy applications.

References

1. W. W. Parson, Modern Optical spectroscopy (2009), Springer Berlin Heidelberg
2. Laser spectroscopy 2nd Edition by W. Demtroder.
3. An Introduction to Laser & their Applications O'Shea, Callen & Rhodes

Phys 675 Laser Physics I (3)

Classical dispersion theory, Classical theory of absorption, Emission, Absorption and Rate Equations, Semi-classical radiation theory, Laser oscillation: Gain, Threshold, Power and frequency, Multimode and transient oscillation, Lasers and pumping mechanisms, Laser resonator and optical coherence, Amplification in Laser media, amplification conditions, Lorentzian line, shapes, Gaussian line, shapes, simple cavity model, Fabry Perot cavity, and Laser Modes, laser gain conditions, Four level system, Mode locking in multimode laser, mode lock in laser, active mode locking, passive mode locking, the Kerr lens, Gas Lasers, laser media, Q-switching, titanium sapphire laser, neodymium YAG and glass lasers, other rare earth lasers, summary of other laser types.

References

1.Laser by Milonni and Eberly.

2. An Introduction to Laser & their Applications O'Shea, Callen & Rhodes

2. Introduction to Laser Physics K. Shimoda

3. Laser Physics M. Sargent, M. O. Scully & W. E. Lamb

4. Lasers Siegman

5. Lasers Svelto

Phys 676 Cavity Mode and Atom Dynamics (3)

Cavity mode dynamics: The master equation, The Fokker-Planck equation, The quantum Langevin equation, Input-output relation; Two-level atom dynamics: The interaction Hamiltonian, Collapse and revival, The master equation, Power spectrum.

References

- Fesseha Kassahun, Fundamentals of Quantum Optics (Lulu, Raleigh, NC, 2008).
- M.O. Scully and M.S. Zubairy, Quantum Optics (Cambridge University Press, Cambridge, 1997).
- D.F. Walls and G.J. Milburn, Quantum Optics (Springer-Verlag, Berlin, 1994).

Phys 677 Resonance Fluorescence and Laser Dynamics (3)

Resonance fluorescence in open space and in a cavity: Atomic expectation values, Power spectrum,Photon antibunching, Photon statistics; Laser dynamics: Fokker-Planck equation, Photon statistics,Spectrum of intensity fluctuations, Power spectrum

References

- Fesseha Kassahun, Fundamentals of Quantum Optics (Lulu, Raleigh, NC, 2008).
- M.O. Scully and M.S. Zubairy, Quantum Optics, (Cambridge University Press,

Cambridge, 1997).

- D.F. Walls and G.J. Milburn, Quantum Optics (Springer-Verlag, Berlin, 1994).

Phys 678 Laser Physics II (3)

Laser Hardware, wavelength dependence, semiconductor lasers, Steady state gain dynamics, Resonator Physics, Electric field, distribution in cavities, geometric optics applied to cavities, cavity stability conditions, Gaussian Beam, divergence angle and beam parameters, beam waist and Rayleigh region, Gaussian beams in cavities, transformation of Gaussian beam, mode matching, Coherent and chaotic states, The quantum radiation field, Density matrix, Laser Photon statistics, The Fokker- Plank equation, Laser Power spectrum.

References

1.Laser by Milonni and Eberly.

2. An Introduction to Laser & their Applications O'Shea, Callen & Rhodes

2. Introduction to Laser Physics K. Shimoda

3. Laser Physics M. Sargent, M. O. Scully & W. E. Lamb

4. Lasers Siegman

5. Lasers Svelto

Phys 679 Quantum States of Light (3)

The quantum radiation field, Number states, Coherent states, Chaotic states, The P function, The Q function, The photon number distribution, Squeezed states.

References

- Fesseha Kassahun, Fundamentals of Quantum Optics (Lulu, Raleigh, NC, 2008).
- M.O. Scully and M.S. Zubairy, Quantum Optics (Cambridge University Press, Cambridge, 1997).
- D.F. Walls and G.J. Milburn, Quantum Optics (Springer-Verlag, Berlin, 1994).

Phys 681 Nuclear Physics I (3)

Addition of angular momentum: Clebsch-Gordon coefficients, Racah coefficients, Spherical tensors, Winger-Echart theorem with applications, Evidence of saturation property, Exchange character, Spin dependence, Charge independence, Isospin formalism, Central and tensor forces, General form of N-N force, Dipole and quadruple moments of deuteron, S-wave effective range theory, proton-proton scattering, Evidence of hard core potential, Nuclear Shell model, Collective model, Physical concepts of a unified model, Electromagnetic decays, Selection rules, Fermi's theory of beta decay, Kurie plot, Fermi and Gamow- Teller transitions, Parity violation in beta decay, Introduction to nuclear reactions.

References

1. Introductory Nuclear Physics Kenneth S. Krane, John Wiley (1988)
2. Elements of Nuclear Physics W. E. Burcham, Longman (1986)
3. An Introduction to Nuclear Physics W. N. Cottingham and D. A. Greenwood Quantum Electronics YarivJ. M Blatt and V.E. Weisskipf: Theoretical Nuclear Physics
4. B.K. Agrawl : Nuclear Physics, Lokbharti Pub, Allahabad. 1989
5. RM. Singru : Introductory Experimental Nuclear Physics
6. H. Enge -Introduction to Nuclear Physics, Addition-Wesley, 1970
7. Kaplan - Nuclear Physics, Addison Wesley, 1963

Phys 682 Nuclear Physics II (3)

Kinematics of nuclear reactions, Channel quantum numbers, Elastic and inelastic processes, Kramer-Kroning dispersion relation, R-matrix theory, Compound nucleus model, Level density, Theory of Direct reactions, Deuteron stripping and pickup reactions, Theory of meson-nucleon scattering, Heavy-ion reactions, Multipole expansion, Angular distribution of multiple radiation, Parity considerations and selection rules, Gamma decay transition probability and its quantum mechanical derivation.

References

1. Introductory Nuclear Physics Kenneth S. Krane, John Wiley (1988)
2. Elements of Nuclear Physics W. E. Burcham, Longman (1986)
3. An Introduction to Nuclear Physics W. N. Cottingham and D. A. Greenwood Quantum Electronics YarivJ. M Blatt and V.E. Weisskipf: Theoretical Nuclear Physics
4. B.K. Agrawl : Nuclear Physics, Lokbharti Pub, Allahabad. 1989
5. RM. Singru : Introductory Experimental Nuclear Physics
6. H. Enge -Introduction to Nuclear Physics, Addition-Wesley, 1970
7. Kaplan - Nuclear Physics, Addison Wesley, 1963

Phys 683 Atomic and molecular physics (3)

Quantum theory of atomic structure; fine structure of spectral lines; Lamb shift; Zeeman effect; Paschan- Back effect; Helium atom; Thomas – Fermi model; Hartree- Fock approximation; Rotational and vibration spectra of molecules; electronic bands; Raman effect and its theory; application of raman effect; Heitler- London theory of hydrogen molecule.

References

1. Atomic spectra and atomic structure - G. Hertzberg
2. Introduction to atomic spectra -H. white
3. Atomic and nuclear physics (vol I) - S.N. Ghoshal
4. Physics of atoms and molecules _ B.H Bransden and c.J. Jachain
5. Quantum theory of atomic structure (vol II) - J.c. Slater

Phys 684 Biological Physics/Biophysics

Biomolecular simulations, biomolecular structure and modeling, foundation of molecular mechanics in terms of force field, composition and evaluation techniques, simulation techniques and their practical application in protein modeling.

Phys 685 Radiological Physics I (3)

construction and operation of x-ray tubes, x-ray tubes and rectifiers, diagnostic x-ray tubes, monitoring and protection of x-ray tubes, atomic physics, electromagnetic radiations, x-rays and matter, the production of x-rays, factors affecting x-ray beam quality and quantity, interaction of x-rays with matter, the radiographic image, geometric radiography, experimental error, the inverse square law

Phys 686 Radiological Physics II (3)

Characteristic of sound, Interaction of ultrasound with matter, Transducers, Beam properties, Image data acquisition, Two dimensional image display and storage, Image quality and artifacts, Doppler Ultrasound: Magnetization properties Generation and detection of the magnetic resonance signal Pulse sequenced Inversion recovery, Gradient recalled Echo Artifacts, Instrumentation, Geometric Tomography, Computed tomography (basic principle), Geometry and Historical development, Detectors and detector arranges, Detection of Acquisition Tomography , Radiation dose and image quality, Artifacts

References

1. Diagnostic Ultrasound, Principles and instrumentation
   Fifth edition, By Frederlok W. Kremkau
2. The essencial physics of medical imaging, second edition, By Jerrol

Phys 691 Seminar in Physics (1)

Approaches in scientific writing, elements of scientific paper, seminar writing and presentation based on sound scientific paper.

Phys 692 Research Methodology in physics

nature and characteristics of research, research problems and objectives, review of related literature, research design, qualities of a good research instrument, sampling designs, data processing and statistical treatment, data analysis and interpretation, form and style in writing a research (thesis and proposal), Project Work.

References

1. Paler-Calmorin, Laurentina. Methods of Research and Thesis Writing, 2006. .
2. Rex Bookstore, Inc. Manila, Philippines Temechegn Engida. Educational Research Methods (Module), 2008.
3. Louis Cohen, Lawrence Manion and Keith Morrison. Research Methods in Education 5th ed.,. Routledge Falmer, London, 2000.
4. Judith Bell. Doing Your Research Project (3rd Edition). Open University Press, United Kingdom, 1999.
5. Joseph Gibaldi. MLA Handbook for Writers of Research Paper 6th ed.,. First East-West Press Edition, New Delhi, 2004

Phys 695 Selected Topics (3)

A course in which various selected topics in physics are covered based on the availability of current topics related to the specialization of the graduate.

Phys 696 Graduate Project

The students are expected to review the existing work on the topics agreed on with the supervisor beforehand independently with the aid of the advice and direction from the supervisor. In all the courses, oral presentation by the student is highly expected.

Phys 699 MSc Thesis

A research projects (Theoretical, Experimental or Computational) to be carried out in different areas of physics under the supervision of advisor.

Phys 751 Electronics I (3)

Introduction to Integrated Circuits, Differential and operational amplifiers using Transistors, Negative feedback configurations, Mathematical operations, Non-linear applications, Comparators, Relaxation oscillator, Log and antilog amplifiers, Mathematical circuits, Principle of operation and application circuits, Introduction to Digital logic gates, Combinational circuits, Reduction using Karnaugh map, Implementation using universal gates, Look-ahead carry implementation, Binary BCD addition, Decoders and encoders, Multiplexers and demultiplexers their applications, Flip-flops, Types and implementation, Conversions, Triggering, Master/slave implementation, Registers, Binary up down counters, Synchronous counters, Ring and Johnson counters, Random sequence generators, Applications of digital circuits in digital clock, Digital voltmeter.

Phys 752 Electronics II (3)

Phase-locked loop and its applications, Filters, Power and Tuned Amplifiers, Analog Multiplier and its applications, Instrumentation Amplifiers, Sensors and Transducers, Telemetry, Noise reduction techniques, Frequency analysis using software packages such as MATLAB.

Phys 753 Physical Electronics I (3)

Operational Amplifier, Frequency response, large signal effects, Integrators and Differentiators, Feedback amplifiers, Feedback configurations, Series-series, Shunt-series loop gain, stability, frequency response, effect of feedback on Amplifier poles, frequency compensation, Single stage Integrated Circuit Amplifiers, steering active load amplifiers, miller effect and frequency response, MOS and BJT cascade amplifiers, source and emitter followers .

References

1. Bernard Grob, Basic Electronics, 4th ed., McGraw Hill International Book Company, London, (1983).
2. Sze S. M.: Semiconductor devices - Physics and technology, John Wiley & Sons, New York 1985.
3. Tayal D. C. Basic Electronics, 2nd ed., Himalaya Publishing House Mumbai, (1998).
4. Frederick F. Driscoll, Robert F. Coughlin. Solid State devices and Applications, D. B Taraporevala Sons and Co. Pvt Ltd (1981).

5. Close K. J and J Yarwood. Experimental Electronics for Students, London Chapman and Hall (1979).
6. Theraja B.L., R.S. Sedha. Principles of Electronic Devices and Circuits, S. Chand and Company Ltd, New Delhi, (2004).
7. Noel M. Morriss. Semiconductor Devices, Mac Millan Publishers Ltd. (1984).

Phys 754 Physical Electronics II (3)

Differential and Multistage Amplifiers, MOS Differential pair signal operation, common mode gain, CMRR, BJT OP Amps, The 741 Folded cascade CMOS OP Amp Transfer functions, Butterworth and Chebyshev filters, Inductor replacement, All pass circuits, Two Integrator loop topology, Bi- Quad- Active Filters Switched capacitor filter.

References

1. A. E. Fitzgerald, Basic Electrical Engineering.
2. S. M. Sze, Semiconductor devices - Physics and technology
3. R. L. Havill,Elements of Electronics for physical scientists.
4. J. J. Brophy, Basic Electronics for scientists.
5. G. Bernard, Basic Electronics.
6. D. C. Taya,l Basic Electronics.

Phys 767 Fiber Optics and Nonlinear Effects (3)

Propagation of electromagnetic waves through crystals, Faraday, Pockel and Kerr effects and their applications, John, Stoke and Muller matrices, Single and multimode propagation in optical fibers, Scattering of light, Raman effect, Three and four wave mixing, Laser Raman spectroscopy

References

- Lasers by Milonni and Eberly.
- Principles of Lasers 3rd Edition by Svelto.

- Laser spectroscopy 2nd Edition by W. Demtroder.
- Introduction to Fiber Optics by A. Ghatak and K. Tyagrajan.

PhEd 603 Advanced Methods of Teaching Physics

The Nature and Philosophy of Physics, Categories of Philosophy, Educational Philosophy and their Implications for Learning Theories and Physics Education, Models, Methods and Instructional Design in Physics, The Role of Practical Work and Technology in Physics Education, Unit and Lesson Planning in Secondary Physics Education, Assessment and Evaluation in Physics Education.

References

- American Association for the advancement of science(1993). Benchmarks for Science Literacy.
- Barbara et al.(2001). The power of problem-based learning. The natural science teachers association.
- Baybee, R.W.(Ed.)(2002). Learning Science and the science of learning. The natural science teachers association.
- Gott, R. and Duggan, S.(1995). Investigative work in the science curriculum. Open University Press.

PhEd 604 Secondary School Physics Curriculum and Research

Best Practices in International Secondary Physics Education, Issues in Secondary Physics Curriculum, Dealing with National Secondary Physics Curriculum, Implementation of Secondary Physics Curriculum, Research on Ways of Alleviating Secondary School-based Problems,

References

- American Association for the advancement of science (1993). Benchmarks for Science Literacy.

- Barbara et al.(2001). The power of problem-based learning. The natural science teachers association.
- Baybee, R.W.(Ed.)(2002). Learning Science and the science of learning(9th edition). The natural science teachers association.
- Bell, R. et al.(2008). Technology in the secondary science classroom. Arlington, VA: NSTA press.
- Commonwealth of Virginia(2003). Standards of learning for Virginia public schools.
- MacNeil, J.D.(1996). Curriculum: A Comprehensive introduction. John Wiley & Sons, Inc.
- Martin,D.J.(2003). Elementary science methods: A constructivist approach(3rd edition). Wadsworth/ Inc.
- Ornstein, A.C. & Hunkins, F.P. (2004). Curriculum foundations, principles, and issues(4th edition). Pearson, Education Inc.
- Wellington, J.J.(2006). Teaching and learning secondary science: Contemporary issues and practical approaches. Routledge, London.